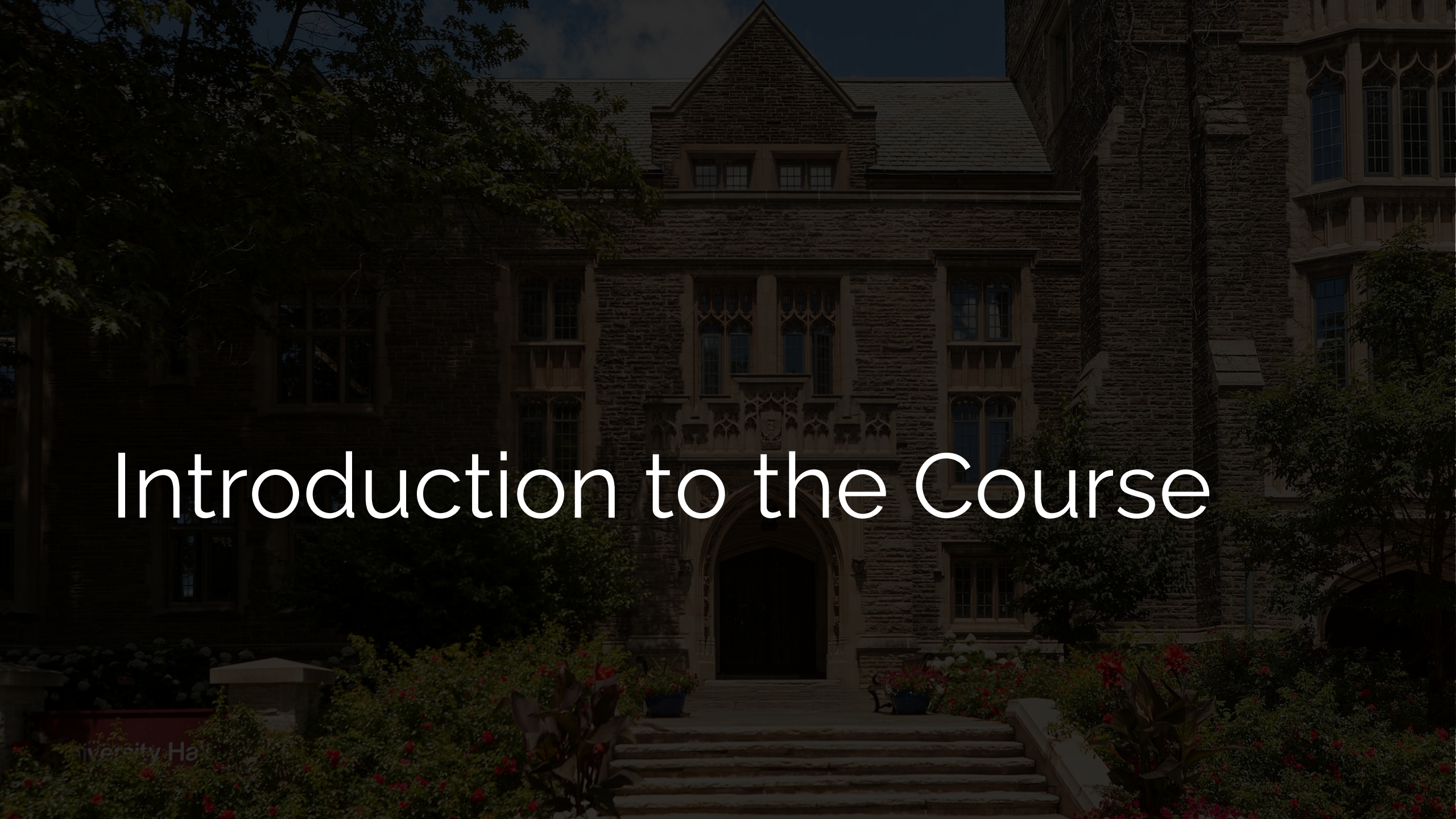
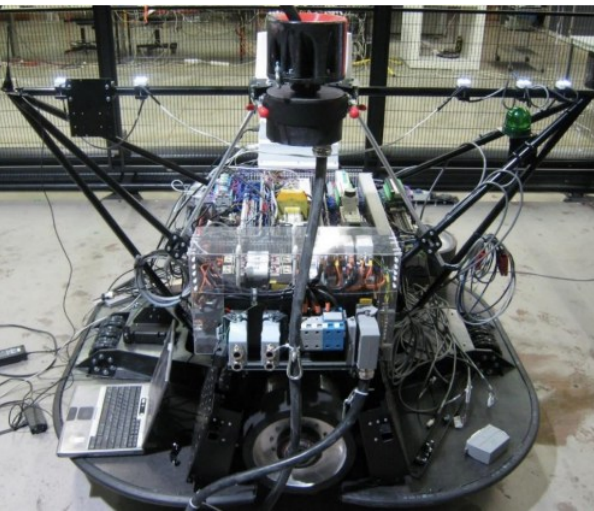




Introduction to the Course

About me

- Instructor: Dr Istvan David
- Office: ITB 160
- Office hours:
 - By appointment (email)
 - Subject: "[2AA4] ..."
 - Prefer Mondays and Thursdays
- Email: istvan.david@mcmaster.ca
- Useful info: istvandavid.com



Background

Research  Sustainable Systems and Methods

modeling and simulation

digital twins and C(B)PS

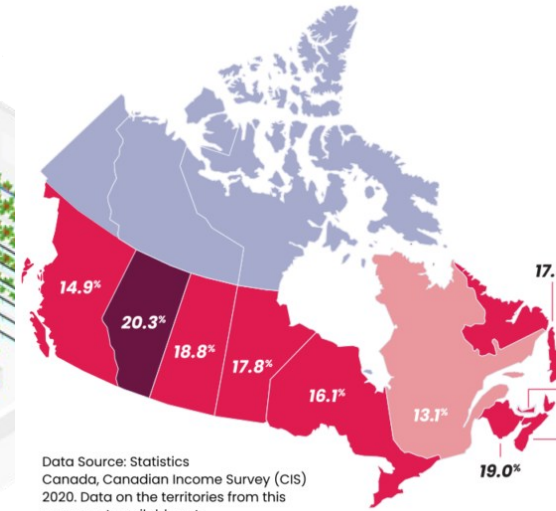
reinforcement learning

sustainable systems

green software and AI



Head of Innovation
R&D Software Engineer
Software Developer/Architect



On the note of sustainability

ICT carbon footprint

Currently:

Estimated by 2040:

Required carbon decrease:

By 2030:

By 2040:

By 2050:

What produces carbon?

AI, big data, blockchain... any computation, really.

=



It is estimated that emails sent globally can contribute to 150M tons of CO₂ emissions (2019), or about 0.3% of the world's carbon footprint.



Before sending it, ask yourself:
do I *really* need to send this email?
Don't spam – save some trees.

Course Organization

- **Lectures:** 3 per week, in person
- **Tutorials:** 1 per week (for each tutorial group), in person, starting today!



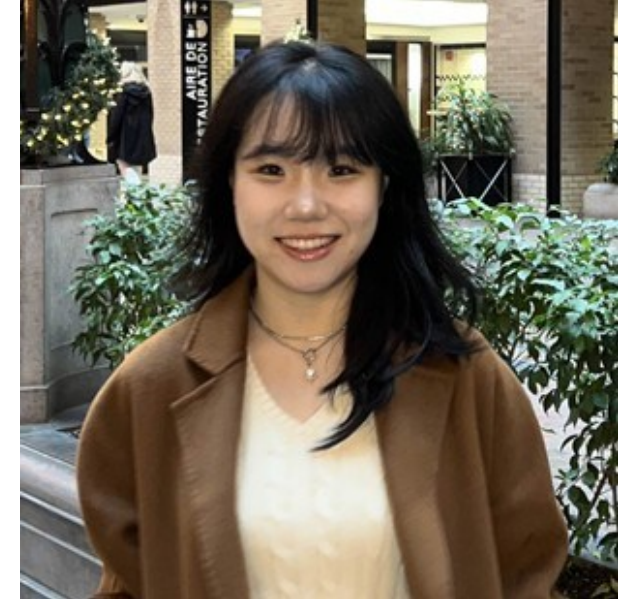
Richard Li
(Mon T)



Aditya Tripuraneni
(Tue T)



Eden Nalian
(Fri T)



Sharon Liu
(Assignments)

• Evaluation

- 2 individual assignments – worth 10% each
- 1 group assignment – worth 15%
- 50-minute midterm worth 25%
- Final exam worth 40%

A1: Jan 13 – Feb 3
A2: Feb 3 – Mar 10
M: Feb 13
A3: Mar 10 – Mar 24

Prerequisites for 2AA4

- SFWRENG 2DM3: Discrete Mathematics with Applications I
 - Set theory, operators, relations (equivalence/identity), substitutions
- SFWRENG 2OP3: Object-oriented programming
 - Classes, interfaces, visibility, message sending, static ≠ instance
 - Java programming ecosystem
- SFWRENG 2XC3: Practice & Experience - Development Basics
 - Unix/Linux environment, scripting, version control (T01)
- Also assumed: you know how to maintain a healthy development environment on your computer.

Topics organized by weeks

1. From Coding to Development
2. Organizing your code
3. Building the product right
4. Contracts & Responsibilities
5. Building the right product
6. Algebraic foundations
7. [READING WEEK]
8. Designing with polymorphism
9. Case study
10. The need for patterns
11. Patterns in action
12. Case study
13. Metrics & Visualization

(Tentative schedule.)

Tutorials **complement** lectures

1. From Coding to Development
2. Organizing your code
3. Building the product right
4. Contracts & Responsibilities
5. Building the right product
6. Algebraic foundations
7. [READING WEEK]
8. Designing with polymorphism
9. Case study
10. The need for patterns
11. Patterns in action
12. Case study
13. Metrics & Visualization

(Tentative schedule.)

1. Engineering Software
2. Vertical Feature Decomposition
3. Module Interface Specification
4. Modelling with the UML (structure)
5. Unit Testing
6. Modelling with the UML (behavior)
7. [READING WEEK]
8. Code Quality
9. Case study
10. Case study
11. Case study
12. Analysis of Patterns
13. Case study

(Tentative topics.)

Tutorials **complement** lectures

1. From Coding to Development
2. Organizing your code
3. Building the product right

1. Engineering Software
2. Vertical Feature Decomposition
3. Module Interface Specification

Tutorials cover material that is not 100% covered in the lectures.

Yes, tutorial content might be in the final exam.

11. Patterns in action
12. Case study: Pure OO design
13. Metrics & Visualization

(Tentative schedule.)

11. Case study: Arithmetic expressions
12. Analysis of Patterns
13. Case study: Poker hands

(Tentative topics.)

This is a transformative course

Leadership roles, C-suite, entrepreneurship...

...

Software architect



More field experience and continued self-development

Actual software e.n.g.i.n.e.e.r.



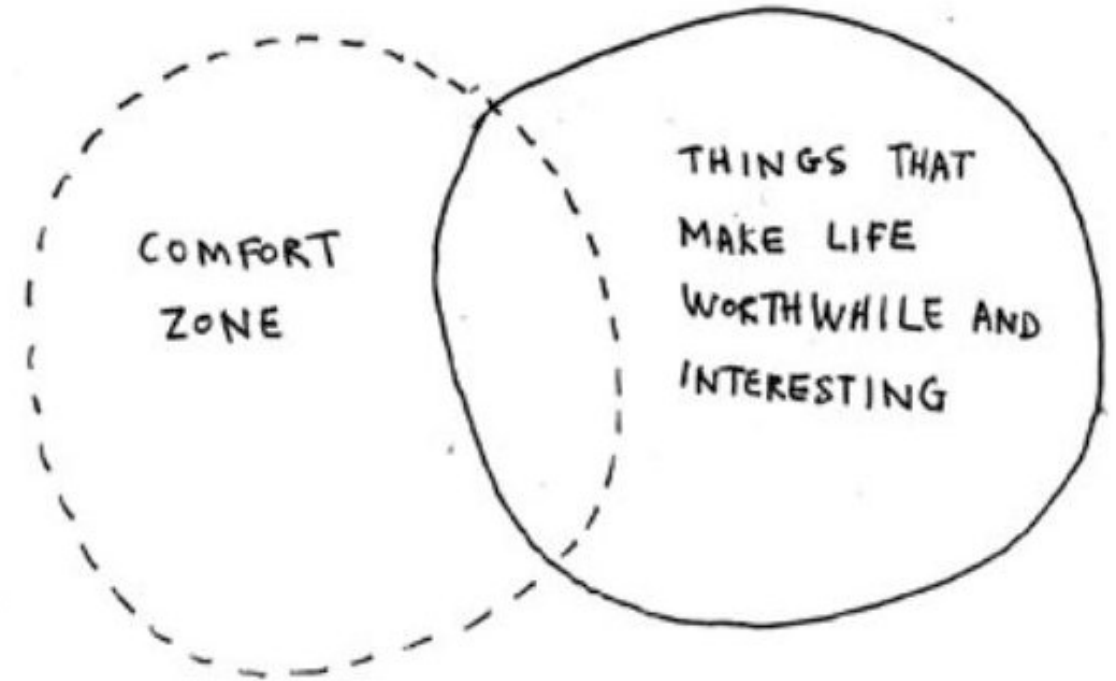
Field experience

Software engineer



2AA4, coop

Programmer



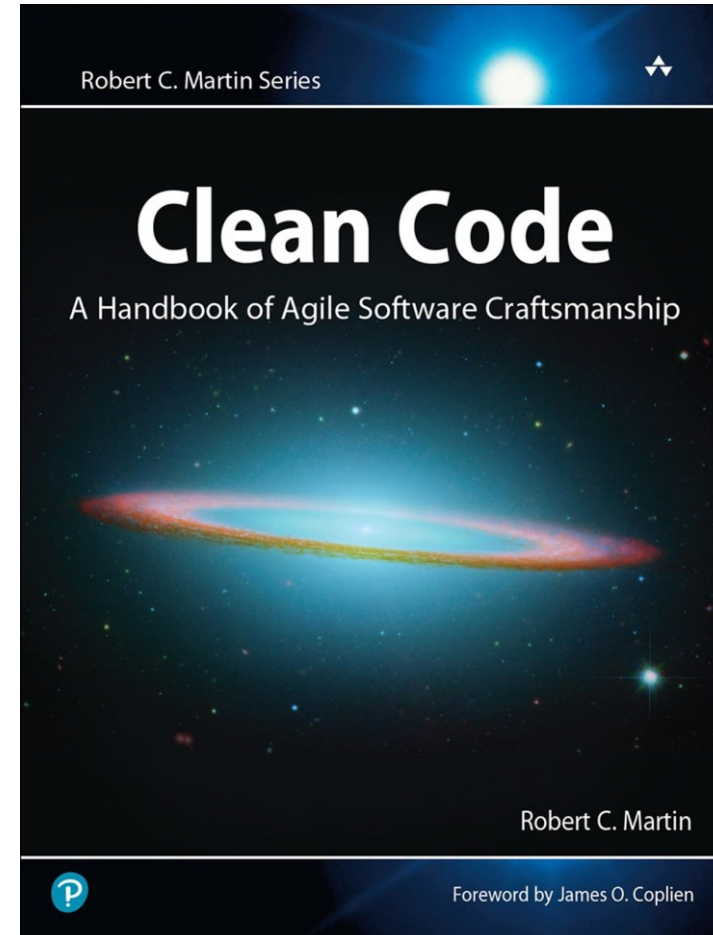
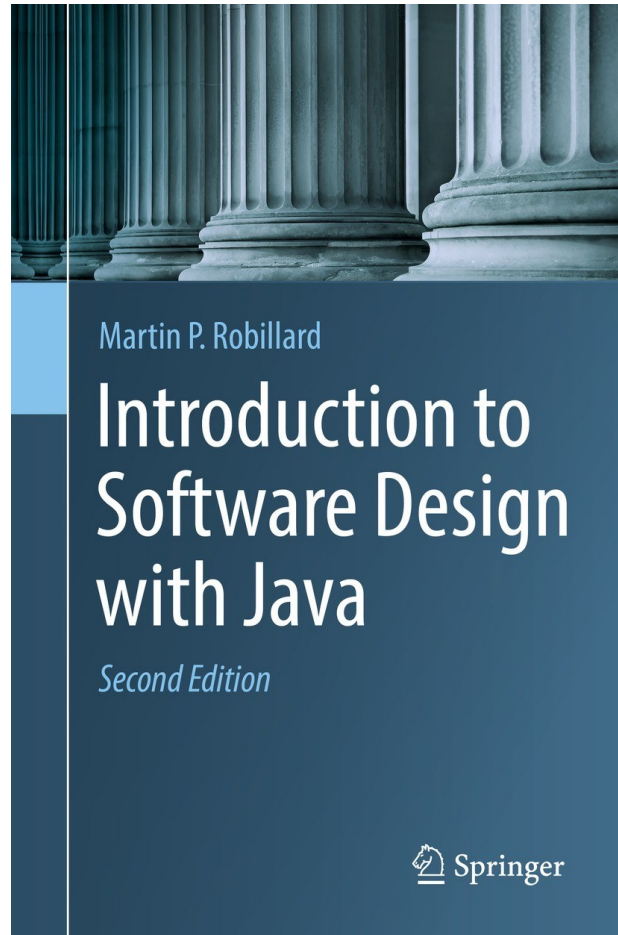
Success is proportional to the effort.

Effort needed in 2AA4?

This is a four-unit course.

Expect 30% more and harder work than in three-unit courses.

Recommended textbooks



Might sample from these in the course.

Academic dishonesty

Zero-tolerance.

If at least one person cheats and the cheater is not held accountable, the Nash equilibrium of the system is:
everyone will cheat to win; and those who don't cheat will lose.

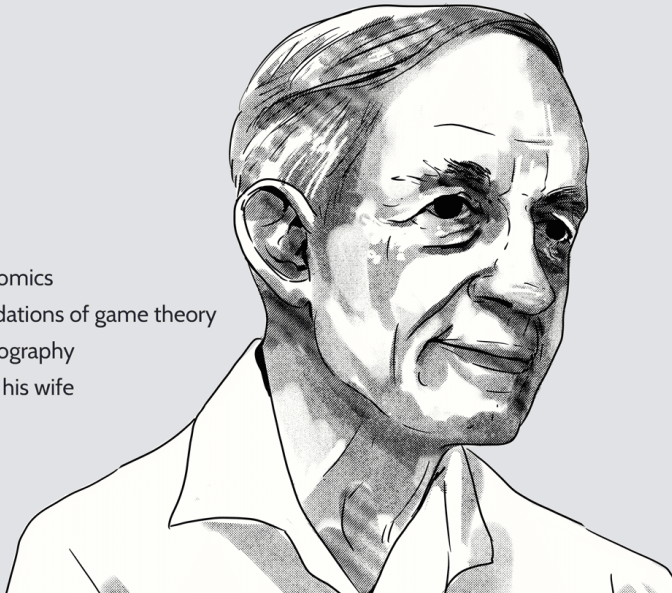
John F. Nash, Jr.

Born: June 13, 1928

Died: May 23, 2015

Mathematician

- 1994 Nobel Prize Recipient in Economics
- Developed the mathematical foundations of game theory
- *A Beautiful Life*, a film based on a biography by Sylvia Nasar, chronicles life with his wife and struggles with mental illness



Engineering design (CEAB)

“ Engineering design is a process of making informed decisions to creatively devise products, systems, components, or processes to meet specified goals based on engineering analysis and judgement. The process is often characterized as complex, open-ended, iterative, and multidisciplinary. Solutions incorporate natural sciences, mathematics, and engineering science, using systematic and current best practices to satisfy defined objectives within identified requirements, criteria and constraints. Constraints to be considered may include (but are not limited to): health and safety, sustainability, environmental,

ethical

, security, economic, aesthetics and human

factors, feasibility and compliance with regulatory aspects, along with universal design issues such as

societal

, cultural and diversification facets.



https://engineerscanada.ca/sites/default/files/2021-06/CEAB%20engineering%20design%20consultation%20report_Final.pdf

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Engineering design (CEAB)

“

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